Find the Error

Grade 3 · 3.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

There are 4 bags with 6 apples each. How many apples? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 4 and 6. The bags are equal groups, so multiply 4 × 6 = 24.**

Correct answer: **24 apples**

Find the Error

Grade 3 · 3.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

There are 3 rows with 8 chairs each. How many chairs? Student's answer: 11

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added instead of multiplying. 3 rows of 8 is 3 × 8 = 24.**

Correct answer: **24 chairs**

Find the Error

Grade 3 · 3.OA.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write 5 groups of 7 as a multiplication equation. Student writes: 5 + 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student wrote an addition. Five groups of seven is 5 × 7.**

Correct answer: **5 × 7 = 35**

Find the Error

Grade 3 · 3.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

24 ÷ 6 = ? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student gave the divisor, not the quotient. 6 × 4 = 24, so 24 ÷ 6 = 4.**

Correct answer: 4

Find the Error

Grade 3 · 3.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

18 stickers are put into groups of 3. How many groups? Student's answer: 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 3 from 18. Divide 18 by 3 to find the groups: 6.**

Correct answer: **6 groups**

Find the Error

Grade 3 · 3.OA.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Share 20 marbles equally among 4 kids. How many each? Student's answer: 16

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted 4 from 20. Divide 20 by 4 to share equally: 5 each.**

Correct answer: **5 marbles each**

Find the Error

Grade 3 · 3.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A box has 8 crayons. There are 6 boxes. How many crayons? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 8 and 6. There are 6 equal boxes, so multiply 6 × 8 = 48.**

Correct answer: **48 crayons**

Find the Error

Grade 3 · 3.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bus carries 36 students shared equally among 4 vans. How many per van? Student's answer: 40

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 36 and 4. The students are shared equally, so divide 36 ÷ 4 = 9.**

Correct answer: **9 students per van**

Find the Error

Grade 3 · 3.OA.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A pack has 9 cards. Lisa has 5 packs. How many cards? Student's answer: 14

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added. Five equal packs means multiplying 9 × 5 = 45.**

Correct answer: **45 cards**

Find the Error

Grade 3 · 3.OA.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

? × 4 = 28. Student's answer: 24

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student found 4 × 6 but not 28. The unknown factor is 28 ÷ 4 = 7.**

Correct answer: 7

Find the Error

Grade 3 · 3.OA.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

6 × ? = 42. Student's answer: 36

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 6 × 6 instead. The unknown factor is 42 ÷ 6 = 7.**

Correct answer: 7

Find the Error

Grade 3 · 3.OA.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

35 ÷ ? = 5. Student's answer: 30

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted incorrectly. 5 × 7 = 35, so the unknown number is 7.**

Correct answer: 7

Find the Error

Grade 3 · 3.OA.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says 3 × 4 is not the same as 4 × 3. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student misused the commutative property. Changing the order of factors does not change the product.**

Correct answer: **No. Both equal 12.**

Find the Error

Grade 3 · 3.OA.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student uses the distributive property: 7 × 6 = 7 × 5 + 1 = 36. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 1 instead of 7 for the second part. 6 = 5 + 1, so 7 × 1 = 7 and 35 + 7 = 42.**

Correct answer: **No. 7 × 6 = 42.**

Find the Error

Grade 3 · 3.OA.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student solves $(2 \times 3) \times 4$ as $2 \times (3 + 4) = 14$. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student changed × to + in the grouping. (2 × 3) × 4 = 6 × 4 = 24.**

Correct answer: **No. (2 × 3) × 4 = 24.**

Find the Error

Grade 3 · 3.OA.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

24 ÷ 8 = ? Student says: 4, because 8 + 4 = 12.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used addition. Division asks what times 8 is 24, and 8 × 3 = 24.**

Correct answer: 3

Find the Error

Grade 3 · 3.OA.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

? ÷ 4 = 6. Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 4 and 6. The unknown number is 6 × 4 = 24.**

Correct answer: 24

Find the Error

Grade 3 · 3.OA.6

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

6 × ? = 30. Student's answer: 36

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 6 × 6. The unknown factor is 30 ÷ 6 = 5.**

Correct answer: 5

Find the Error

Grade 3 · 3.OA.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

7 × 8 = ? Student's answer: 54

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a multiplication error. 7 × 8 = 56.**

Correct answer: **56**

Find the Error

Grade 3 · 3.OA.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

6 × 9 = ? Student's answer: 45

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 5 × 9. 6 × 9 = 54.**

Correct answer: **54**

Find the Error

Grade 3 · 3.OA.7

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

56 ÷ 7 = ? Student's answer: 6

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a division error. 7 × 8 = 56, so 56 ÷ 7 = 8.**

Correct answer: 8

Find the Error

Grade 3 · 3.OA.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A store has 5 boxes with 8 pencils each, then sells 6 pencils. How many are left? Student's answer: 34

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted wrong. 5 × 8 = 40 and 40 + 0 is not what happens here. Add all the pencils first, then take away the sold ones, 40 − 6 = 34.**

Correct answer: **34 pencils (5 × 8 = 40, then 40 − 6)**

Find the Error

Grade 3 · 3.OA.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Jake buys 4 packs of 6 gum and 3 extra pieces. How many pieces in all? Student's answer: 13

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the pack number instead of multiplying. 4 × 6 = 24, and 24 + 3 = 27.**

Correct answer: **27 pieces**

Find the Error

Grade 3 · 3.OA.8

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

24 apples are split evenly into 3 bags. Then 2 more apples go into each bag. How many in each bag? Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student left out the 2 extra apples. 24 ÷ 3 = 8, then 8 + 2 = 10.**

Correct answer: **10 apples**

Find the Error

Grade 3 · 3.OA.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Pattern: 4, 8, 12, 16. Student's next number: 18

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 2 instead of 4. The pattern adds 4 each time, so the next number is 20.**

Correct answer: 20

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Pattern: 3, 6, 9, ... Student's next number: 13

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added 4 instead of 3. The pattern adds 3, so the next number is 12.**

Correct answer: 12

Find the Error

Grade 3 · 3.OA.9

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

2 × 5 = 10, 4 × 5 = 20, 6 × 5 = 30. Student says the next product is 8 × 5 = 45.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student solved 8 × 5 wrong. The pattern adds 10 each time, and 8 × 5 = 40.**

Correct answer: **40**

Find the Error

Grade 3 · 3.NBT.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 46 to the nearest ten. Student's answer: 40

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student rounded down. 46 is closer to 50 than to 40.**

Correct answer: **50**

Find the Error

Grade 3 · 3.NBT.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 268 to the nearest hundred. Student's answer: 200

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student rounded down. 268 is closer to 300 than to 200.**

Correct answer: **300**

Find the Error

Grade 3 · 3.NBT.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Round 85 to the nearest ten. Student's answer: 80

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student rounded down. 85 rounds up to 90.**

Correct answer: **90**

Find the Error

Grade 3 · 3.NBT.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

245 + 128 = ? Student's answer: 363

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made an addition error. 245 + 128 = 373.**

Correct answer: 373

Find the Error

Grade 3 · 3.NBT.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

500 - 275 = ? Student's answer: 325

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made a subtraction error. 500 - 275 = 225.**

Correct answer: **225**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

330 + 470 = ? Student's answer: 700

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student made an addition error. 330 + 470 = 800.**

Correct answer: **800**

Find the Error

Grade 3 · 3.NBT.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

3 × 70 = ? Student's answer: 2,100

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student put the zeros in the wrong place. 3 × 7 = 21, so 3 × 70 = 210.**

Correct answer: **210**

Find the Error

Grade 3 · 3.NBT.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

5 × 30 = ? Student's answer: 35

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student forgot the zero. 5 × 3 = 15, so 5 × 30 = 150.**

Correct answer: **150**

Find the Error

Grade 3 · 3.NBT.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

8 × 40 = ? Student's answer: 3,200

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added an extra zero. 8 × 4 = 32, so 8 × 40 = 320.**

Correct answer: **320**

Find the Error

Grade 3 · 3.NF.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says $\frac{1}{4}$ means 4 parts out of 1. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student reversed the meaning. The bottom number is the number of equal parts in the whole, and $\frac{1}{4}$ means 1 of 4 equal parts.**

Correct answer: **No. $\frac{1}{4}$ is 1 of 4 equal parts.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says $\frac{1}{3}$ of a shape means 3 parts are shaded. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student mixed up the numerator. $\frac{1}{3}$ means 1 of the 3 equal parts is shaded.**

Correct answer: **No. $\frac{1}{3}$ means 1 part out of 3 equal parts.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says the top number of a fraction is the number of equal parts. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The bottom number tells how many equal parts the whole has. The top number tells how many parts are counted.**

Correct answer: **No. The bottom number is the number of equal parts.**

Find the Error

Grade 3 · 3.NF.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Place $\frac{1}{2}$ on a number line from 0 to 1. Student places it at $\frac{1}{4}$.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student placed the fraction at a quarter. $\frac{1}{2}$ is halfway between 0 and 1.**

Correct answer: **Halfway between 0 and 1**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A number line from 0 to 1 has 4 equal parts. Student marks $\frac{3}{4}$ at the 4th mark.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student marked the endpoint. $\frac{3}{4}$ is the third mark from 0.**

Correct answer: **The 3rd mark from 0**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says $\frac{1}{4}$ and $\frac{2}{4}$ are at the same point on a number line.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student thought the two fractions were the same point. $\frac{2}{4}$ is twice as far from 0 as $\frac{1}{4}$.**

Correct answer: **No. $\frac{1}{4}$ and $\frac{2}{4}$ are different points.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $\frac{1}{2}$ equal to $\frac{2}{4}$? Student's answer: no

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student did not see that the fractions are equivalent. $\frac{2}{4}$ is the same amount as $\frac{1}{2}$.**

Correct answer: Yes

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Write a fraction equal to $\frac{2}{3}$ with a denominator of 6. Student's answer: $\frac{4}{3}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student changed only the numerator. Multiply both the top and bottom by 2 to get $\frac{4}{6}$.**

Correct answer: $\frac{4}{6}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Compare $\frac{3}{4}$ and $\frac{3}{8}$. Student writes: $\frac{3}{4} < \frac{3}{8}$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student reversed the comparison. $\frac{3}{4}$ uses bigger pieces, so $\frac{3}{4}$ is greater.**

Correct answer: $\frac{3}{4} > \frac{3}{8}$

Find the Error

Grade 3 · 3.MD.1

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A task starts at 1:20 and ends at 2:05. How many minutes? Student's answer: 85

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the minutes. From 1:20 to 2:00 is 40 minutes, then 5 more is 45.**

Correct answer: **45 minutes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A class starts at 9:30 and ends at 10:00. How many minutes? Student's answer: 70

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added wrong. From 9:30 to 10:00 is 30 minutes.**

Correct answer: **30 minutes**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Start 4:10. End 4:45. How many minutes? Student's answer: 45

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted the end minutes only. From 4:10 to 4:45 is 35 minutes.**

Correct answer: **35 minutes**

Find the Error

Grade 3 · 3.MD.2

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bottle holds 2 liters. Student's answer: 2 milliliters

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the smallest unit. 2 liters is 2,000 milliliters.**

Correct answer: **2,000 milliliters**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bag of rice weighs 1 kilogram. Student's answer: 1 gram

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the smallest mass unit. 1 kilogram is 1,000 grams.**

Correct answer: **1,000 grams**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A paperclip weighs about 1 gram. Student's answer: 1 kilogram

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used a much bigger unit. A paperclip weighs about 1 gram, which is far less than a kilogram.**

Correct answer: **1 gram**

Find the Error

Grade 3 · 3.MD.3

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A picture graph: each picture stands for 5 kids. There are 6 pictures. Student's answer: 6 kids

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted the pictures, not the kids. 6 pictures times 5 is 30.**

Correct answer: **30 kids**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A bar graph has a scale of 2. The bar reaches 4 marks. Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student read the marks, not the value. Each mark is worth 2, so 4 marks is 8.**

Correct answer: 8

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A picture graph: dogs have 7 pictures and cats have 4 pictures. Each picture is 2 pets. How many more dogs than cats? Student's answer: 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student subtracted the pictures and not the pets. Dogs have 14 and cats have 8, so the difference is 6.**

Correct answer: **6 more**

Find the Error

Grade 3 · 3.MD.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line plot shows 2 X's at $\frac{1}{4}$ inch and 3 X's at $\frac{1}{2}$ inch. How many X's in all? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted only the X's at $\frac{1}{4}$. There are 2 + 3 = 5 X's.**

Correct answer: 5

Find the Error

Grade 3 · 3.MD.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A line plot has 4 X's at $\frac{1}{2}$ inch. Student says the total length is 4 inches.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student multiplied wrong. 4 × $\frac{1}{2}$ = 2 inches.**

Correct answer: **2 inches**

Find the Error

Grade 3 · 3.MD.4

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says $\frac{1}{2}$ inch and $\frac{1}{4}$ inch are the same length.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student thought they were the same. $\frac{1}{2}$ inch is twice as long as $\frac{1}{4}$ inch.**

Correct answer: **No. $\frac{1}{2}$ is longer than $\frac{1}{4}$.**

Find the Error

Grade 3 · 3.MD.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle is covered by 12 unit squares with no gaps. Student's answer: 14 square units

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted wrong. The rectangle is covered by 12 unit squares.**

Correct answer: **12 square units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says the area of a 3 by 4 rectangle is 7 because 3 + 4 = 7.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. Area is found by multiplying: 3 × 4 = 12 square units.**

Correct answer: **12 square units**

Find the Error

Grade 3 · 3.MD.5

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says area and perimeter are the same measurement.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **Area measures the space inside a shape. Perimeter measures the distance around it.**

Correct answer: **No. Area and perimeter are different.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A 3 by 3 region is covered by 9 square tiles. Student says the area is 6 square units.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student missed some squares. The region has 9 square tiles.**

Correct answer: **9 square units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle covers 20 squares that are 1 cm each. Student says 2 square cm.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student counted the rows. The rectangle covers 20 square centimeters.**

Correct answer: **20 square cm**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student measures a rectangle's area with a ruler: 5 cm, and says 5 square cm.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A ruler measures length, not area. Area must be counted with square units.**

Correct answer: **Count the squares that cover the shape.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a rectangle with sides 6 and 4. Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. Area is 6 × 4 = 24.**

Correct answer: **24 square units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a rectangle 7 by 3. Student says: 10 square units because 7 + 3 = 10.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added the sides. Area is 7 × 3 = 21.**

Correct answer: **21 square units**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the area of a rectangle 9 by 2. Student's answer: 18 meters

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the wrong unit. Area is measured in square meters.**

Correct answer: **18 square meters**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the perimeter of a rectangle 5 by 3. Student's answer: 15

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only 2 sides. The perimeter adds all 4 sides: 5 + 3 + 5 + 3 = 16.**

Correct answer: 16

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the perimeter of a square with side 4. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only 2 sides. The perimeter is 4 + 4 + 4 + 4 = 16.**

Correct answer: 16

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle has sides 6, 2, 6, and 2. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student added only 2 sides. The perimeter is 6 + 2 + 6 + 2 = 16.**

Correct answer: 16

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says a square is not a rectangle. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A square has 4 right angles and two pairs of parallel sides, so it is a rectangle.**

Correct answer: **No. A square is a rectangle.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says a rhombus is always a square. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A rhombus has 4 equal sides, but it needs right angles to be a square.**

Correct answer: **No. A rhombus is not always a square.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Student says a trapezoid is a parallelogram. Is this right?

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **A trapezoid has only one pair of parallel sides. A parallelogram has two pairs.**

Correct answer: **No. A trapezoid is not a parallelogram.**

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle is split into 4 parts that are not equal in area. Student says each part is $\frac{1}{4}$.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The parts are not equal in area, so each part is not $\frac{1}{4}$ of the whole.**

Correct answer: **No. The parts must have equal area to be fourths.**

Find the Error

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Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A shape is split into 4 equal parts. Student says each part is $\frac{1}{2}$.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used the wrong denominator. 4 equal parts make fourths, so each part is $\frac{1}{4}$.**

Correct answer: $\frac{1}{4}$

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A rectangle is split into 3 equal parts. Student says each part is $\frac{1}{2}$.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

The error: **The student used 2 instead of 3. 3 equal parts make thirds, so each part is $\frac{1}{3}$.**

Correct answer: $\frac{1}{3}$